

Mod-8 congruence and an interleaving bridge for the horizontal-domino-count residue-class constrained tiling count on the $3 \times n$ board

(Lean 4 machine-checked claim of record)

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Abstract

We record a machine-checked result about two pure integer sequences $a_2, b_2: \mathbb{N} \rightarrow \mathbb{Z}$ defined by an initial segment together with integer-coefficient recurrences: the compressed subcolumn b_2 satisfies the strong congruence $b_2(k) \equiv 1 \pmod{8}$ for every $k \in \mathbb{N}$, the sequence a_2 vanishes at every odd index, and the two *independently defined* sequences agree on the even indices through the interleaving bridge $a_2(2k) = b_2(k)$. All statements are verified by the Lean 4 kernel under the pinned mathlib revision, with no `sorry` and axioms confined to `{propxt, Quot.sound}` (the whitelist also allows `Classical.choice`, which this track does not use). Novelty: search found no occupying record. This note is a priority claim of record, not a full paper.

1 Introduction

This note records a formalized result produced by an AI4Math automated pipeline, in which propositions and proofs are generated by a language model and adjudicated solely by the Lean 4 kernel. The result concerns pure sequence-level objects: an initial-segment recurrence pair (a_2, b_2) , a strong congruence for the compressed subcolumn, and the interleaving identity that

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ties the two definitions together. The note is a priority claim of record, deposited for timestamping; the corresponding narrative paper has been deliberately deferred by the author (see the card), so no full exposition is attempted here. Verification strength is stated in Section 4: strict kernel compilation, zero `sorry`, standard axiom whitelist. The mod-3 object that motivates the sequence is the sum of tiling counts over tilings counted by the number of horizontal dominoes, modulo 3; the kernel layer does not prove that identification, and the sequence is described throughout as a pure integer object. No literature survey is included—that expectation belongs to arXiv moderation and does not apply to a Zenodo deposit.

2 Statement

We first fix notation. Let $a_2: \mathbb{N} \rightarrow \mathbb{Z}$ be the sequence with initial values

$$a_2(0), \dots, a_2(11) = 1, 0, 1, 0, 1, 0, 9, 0, 57, 0, 225, 0$$

and even-lag recurrence

$$a_2(n) = 6a_2(n-2) - 15a_2(n-4) + 28a_2(n-6) - 15a_2(n-8) + 6a_2(n-10) - a_2(n-12),$$

and let $b_2: \mathbb{N} \rightarrow \mathbb{Z}$ be the *independently* defined sequence with initial values

$$b_2(0), \dots, b_2(5) = 1, 1, 1, 9, 57, 225$$

and order-6 recurrence

$$b_2(k) = 6b_2(k-1) - 15b_2(k-2) + 28b_2(k-3) - 15b_2(k-4) + 6b_2(k-5) - b_2(k-6).$$

Neither definition refers to tilings, dominoes or counting; both are pure integer recursions, so the theorems below are structural theorems about sequences. The following three facts are formalized.

Theorem 1 (mod-8 congruence for the compressed subcolumn; main result). *For every $k \in \mathbb{N}$, $b_2(k) \bmod 8 = 1$.*

Theorem 2 (interleaving bridge; main result). *For every $k \in \mathbb{N}$, $a_2(2k) = b_2(k)$.*

Theorem 3 (vanishing at odd indices). *For every $n \in \mathbb{N}$, if $n \bmod 2 = 1$ then $a_2(n) = 0$.*

Listing 1: Frozen formal statement (Lean 4, verbatim from `formalized/02-mod3x3-statements.lean`, lines 38–85), with the three proof-body placeholder lines omitted (see the note below the listing).
sha256 da4eae94...).

```

1  import Mathlib
2
3  /-- **T2 主序列  $a_2$  ( $order-12$  仅偶滞后递推) **.
4  初值  $a_2(0..11) = 1, 0, 1, 0, 1, 0, 9, 0, 57, 0, 225, 0$ ;
5  递推  $a_2(n) = 6 \cdot a_2(-n_2) - 15 \cdot a_2(-n_4) + 28 \cdot a_2(-n_6) - 15 \cdot a_2(-n_8) + 6 \cdot$ 
    $a_2(-n_{10}) - a_2(-n_{12})$ . -/
6  def a2 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
7    | 0 => 1
8    | 1 => 0
9    | 2 => 1
10   | 3 => 0
11   | 4 => 1
12   | 5 => 0
13   | 6 => 9
14   | 7 => 0
15   | 8 => 57
16   | 9 => 0
17   | 10 => 225
18   | 11 => 0
19   | n + 12 =>
20     6 * a2 (n + 10) - 15 * a2 (n + 8) + 28 * a2 (n + 6)
21     - 15 * a2 (n + 4) + 6 * a2 (n + 2) - a2 n
22
23  /-- **T2 压缩子列  $b_2$  (独立定义: 6 个初值 +  $order-6$  递推) **.
24  初值  $b_2(0..5) = 1, 1, 1, 9, 57, 225$ ;
25  递推  $b_2(k) = 6 \cdot b_2(-k_1) - 15 \cdot b_2(-k_2) + 28 \cdot b_2(-k_3) - 15 \cdot b_2(-k_4) + 6 \cdot$ 
    $b_2(-k_5) - b_2(-k_6)$ .
26  (**独立定义**, 非  $a_2(2k)$  的派生——后者会使递推退化为定义展开,
   见文件头返工记录。) -/
27  def b2 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
28    | 0 => 1
29    | 1 => 1
30    | 2 => 1
31    | 3 => 9
32    | 4 => 57
33    | 5 => 225
34    | k + 6 =>
35      6 * b2 (k + 5) - 15 * b2 (k + 4) + 28 * b2 (k + 3)
36      - 15 * b2 (k + 2) + 6 * b2 (k + 1) - b2 k
37

```

```

38 /-- **T2-A (主定理: 强同余) **: 压缩子列模 8 恒为 1。 -/
39 theorem b2_mod8 (k : N) : b2 k % 8 = 1 := by
40
41 /-- **T2-B (交织桥) **: 两个独立定义的序列在偶位处相合——`a2 (2k) =
    b2 k`。
42 这是本轨的**主桥** (两个独立递推定义之间的等式, 需真正归纳, 非定义展
    开)。 -/
43 theorem a2_interleave (k : N) : a2 (2 * k) = b2 k := by
44
45 /-- **T2-C (零方向) **: 主序列的奇数位恒为 0。 -/
46 theorem a2_odd_eq_zero (n : N) (hn : n % 2 = 1) : a2 n = 0 := by

```

This is lines 38–85 of the frozen statement snapshot with exactly three lines removed: the three proof-body placeholder lines (one per theorem). That single, stated elision is the deposit convention (see the sibling claims): the frozen file’s bodies are the formalization department’s placeholders, and reproducing an unfinished-proof marker in this note would be rejected by the deposit’s static lint. Everything else—doc-comments including their internal notes, definitions, binders, signatures and the `:= by` keyword—is verbatim and in order. `audit/inspect-tex.py` checks this mechanically: it asserts that every remaining line occurs in the source in order, and that the only lines missing from the block are those placeholder bodies. The real, sorry-free proofs follow in excerpt.

3 Proof

The complete machine proof is 139 lines long and therefore exceeds a comfortable page; the listing below is a *contiguous excerpt*, lines 36–139 of `final/02-mod3x3-proved.lean`, which covers the whole development: the two definitions, the main congruence Theorem 1 with its full proof body, the interleaving bridge Theorem 2 with its full proof body, and the odd-index vanishing Theorem 3 with its full proof body. The only material left out is lines 1–35 of the file: the header comment block (1–34), which is project metadata rather than mathematical content, and one blank line. The full text, header included, is in the deposit under `proofs/02-mod3x3-proved.lean`. A prose reconstruction is not an obligation of this note: narrative proofs belong to a full paper, which has been deferred.

Listing 2: Verbatim excerpt, lines 36–139 of `final/02-mod3x3-proved.lean` (139 lines, sha256 79c19719...); the

full text is in the deposit's proofs/.

```

1  import Mathlib
2
3  /-- **T2 主序列  $a_2$  (order-12 仅偶滞后递推) **.
4  初值  $a_2(0..11) = 1, 0, 1, 0, 1, 0, 9, 0, 57, 0, 225, 0$ ;
5  递推  $a_2(n) = 6 \cdot a_2(-n2) - 15 \cdot a_2(-n4) + 28 \cdot a_2(-n6) - 15 \cdot a_2(-n8) + 6 \cdot$ 
    $a_2(-n10) - a_2(-n12)$ . -/
6  def a2 : ℕ → ℤ
7    | 0 => 1
8    | 1 => 0
9    | 2 => 1
10   | 3 => 0
11   | 4 => 1
12   | 5 => 0
13   | 6 => 9
14   | 7 => 0
15   | 8 => 57
16   | 9 => 0
17   | 10 => 225
18   | 11 => 0
19   | n + 12 =>
20     6 * a2 (n + 10) - 15 * a2 (n + 8) + 28 * a2 (n + 6)
21     - 15 * a2 (n + 4) + 6 * a2 (n + 2) - a2 n
22
23  /-- **T2 压缩子列  $b_2$  (独立定义: 6 个初值 + order-6 递推) **.
24  初值  $b_2(0..5) = 1, 1, 1, 9, 57, 225$ ;
25  递推  $b_2(k) = 6 \cdot b_2(-k1) - 15 \cdot b_2(-k2) + 28 \cdot b_2(-k3) - 15 \cdot b_2(-k4) + 6 \cdot$ 
    $b_2(-k5) - b_2(-k6)$ .
26  (**独立定义**, 非  $a_2(2k)$  的派生——后者会使递推退化为定义展开,
   见文件头返工记录。) -/
27  def b2 : ℕ → ℤ
28    | 0 => 1
29    | 1 => 1
30    | 2 => 1
31    | 3 => 9
32    | 4 => 57
33    | 5 => 225
34    | k + 6 =>
35     6 * b2 (k + 5) - 15 * b2 (k + 4) + 28 * b2 (k + 3)
36     - 15 * b2 (k + 2) + 6 * b2 (k + 1) - b2 k
37
38  /-- **T2-A (主定理: 强同余) **: 压缩子列模 8 恒为 1. -/
39  theorem b2_mod8 (k : ℕ) : b2 k % 8 = 1 := by
40    refine Nat.strong_induction_on k (fun k ih => ?_)

```

```

41 by_cases hlt : k < 6
42 · interval_cases k <;> norm_num [b2]
43 · obtain ⟨j, rfl⟩ : ∃ j, k = j + 6 := ⟨k - 6, by omega⟩
44   have h5 : b2 (j + 5) % 8 = 1 := ih (j + 5) (by omega)
45   have h4 : b2 (j + 4) % 8 = 1 := ih (j + 4) (by omega)
46   have h3 : b2 (j + 3) % 8 = 1 := ih (j + 3) (by omega)
47   have h2 : b2 (j + 2) % 8 = 1 := ih (j + 2) (by omega)
48   have h1 : b2 (j + 1) % 8 = 1 := ih (j + 1) (by omega)
49   have h0 : b2 j % 8 = 1 := ih j (by omega)
50   rw [show b2 (j + 6) = 6 * b2 (j + 5) - 15 * b2 (j + 4) + 28 *
51       b2 (j + 3)
52         - 15 * b2 (j + 2) + 6 * b2 (j + 1) - b2 j from by simp
53         only [b2]]
54   omega
55 /-- **T2-B (交织桥)**: 两个独立定义的序列在偶位处相合——`a2 (2k) =
56    b2 k`。
57 这是本轨的**主桥**（两个独立递推定义之间的等式，需真正归纳，非定义展
58 开）。 -/
59 theorem a2_interleave (k : ℕ) : a2 (2 * k) = b2 k := by
60   refine Nat.strong_induction_on k (fun k ih => ?_)
61   by_cases hlt : k < 6
62   · interval_cases k <;> decide
63   · obtain ⟨j, rfl⟩ : ∃ j, k = j + 6 := ⟨k - 6, by omega⟩
64     have e5 : a2 (2 * j + 10) = b2 (j + 5) := by
65       rw [show 2 * j + 10 = 2 * (j + 5) by ring]
66       exact ih (j + 5) (by omega)
67     have e4 : a2 (2 * j + 8) = b2 (j + 4) := by
68       rw [show 2 * j + 8 = 2 * (j + 4) by ring]
69       exact ih (j + 4) (by omega)
70     have e3 : a2 (2 * j + 6) = b2 (j + 3) := by
71       rw [show 2 * j + 6 = 2 * (j + 3) by ring]
72       exact ih (j + 3) (by omega)
73     have e2 : a2 (2 * j + 4) = b2 (j + 2) := by
74       rw [show 2 * j + 4 = 2 * (j + 2) by ring]
75       exact ih (j + 2) (by omega)
76     have e1 : a2 (2 * j + 2) = b2 (j + 1) := by
77       rw [show 2 * j + 2 = 2 * (j + 1) by ring]
78       exact ih (j + 1) (by omega)
79     have e0 : a2 (2 * j) = b2 j := ih j (by omega)
80     rw [show 2 * (j + 6) = 2 * j + 12 by ring]
81     rw [show a2 (2 * j + 12) = 6 * a2 (2 * j + 10) - 15 * a2 (2 *
82         j + 8) + 28 * a2 (2 * j + 6)
83           - 15 * a2 (2 * j + 4) + 6 * a2 (2 * j + 2) - a2 (2 * j)
84         from by simp only [a2]]

```

```

80   rw [e5, e4, e3, e2, e1, e0]
81   simp only [b2]
82
83 /-- **T2-C (零方向) **: 主序列的奇数位恒为 0。 -/
84 theorem a2_odd_eq_zero (n : ℕ) (hn : n % 2 = 1) : a2 n = 0 := by
85   revert hn
86   refine Nat.strong_induction_on n (fun n ih hn => ?_)
87   by_cases hlt : n < 12
88   · interval_cases n <;> simp_all [a2]
89   · obtain ⟨m, rfl⟩ : ∃ m, n = m + 12 := ⟨n - 12, by omega⟩
90     have h1 : (m + 10) % 2 = 1 := by omega
91     have h2 : (m + 8) % 2 = 1 := by omega
92     have h3 : (m + 6) % 2 = 1 := by omega
93     have h4 : (m + 4) % 2 = 1 := by omega
94     have h5 : (m + 2) % 2 = 1 := by omega
95     have h6 : m % 2 = 1 := by omega
96     have e1 : a2 (m + 10) = 0 := ih (m + 10) (by omega) h1
97     have e2 : a2 (m + 8) = 0 := ih (m + 8) (by omega) h2
98     have e3 : a2 (m + 6) = 0 := ih (m + 6) (by omega) h3
99     have e4 : a2 (m + 4) = 0 := ih (m + 4) (by omega) h4
100    have e5 : a2 (m + 2) = 0 := ih (m + 2) (by omega) h5
101    have e6 : a2 m = 0 := ih m (by omega) h6
102    simp only [a2]
103    rw [e1, e2, e3, e4, e5, e6]
104    ring

```

4 Verification evidence

- Toolchain: Lean 4 v4.34.0; mathlib4 v4.34.0, revision 5ed29652.
- Statement integrity: formalized/02-mod3x3-statements.lean, 86 lines, sha256 da4eae94971f8a5e8f1d74f973183b1643158bb7ff96e88db97a09832db50901. Final artifact final/02-mod3x3-proved.lean, 139 lines, sha256 79c19719835ac4ccbffcc72707fd81dde2e52d32b8e55a4fe89dad1341c70901. Every listing above is byte-identical to its source (checked by audit/inspect-tex.py, which asserts that each listed line occurs in its source in order and that the only omitted lines are the statement file's placeholder proof bodies).
- Kernel check: strict compilation exits 0 with no sorry and no admit. #print axioms output, verbatim:


```
'b2_mod8' depends on axioms: [propext, Quot.sound]
```

```
'a2_interleave' depends on axioms: [propext, Quot.sound]
'a2_odd_eq_zero' depends on axioms: [propext, Quot.sound]
```

All three are subsets of `{propext, Quot.sound, Classical.choice}`; this track does not use `Classical.choice` at all, and `sorryAx` does not occur.

- Audit chain: statement freeze (sha256), byte-wise symmetric-difference comparison against the frozen snapshot, strict-mode compilation, and the axiom whitelist check. Originals are in the deposit under `audit/`.
- Gate-2 rework (recorded because it changed the theorem set). The first frozen version defined `b2` as the derived subcolumn `a2 (2 * k)`. Because `a2` itself is a recurrence with even lags only, the then-listed theorem asserting the order-6 recurrence of `b2` reduced to a definitional unfolding, and the gate-2 bare probes `rfl`, `trivial` and `aesop` each closed it in one tactic: the pipeline’s degeneracy criterion flagged it as a degenerate proposition with no mathematical content. The rework made `b2` an independent definition (six initial values plus its own order-6 recurrence), withdrew the derived recurrence theorem—it is now the definition of `b2`, not a theorem—and added Theorem 2, the interleaving bridge between two independent definitions, which needs genuine induction and is not a one-line `simp`. After the rework the bare probes return zero passes.

5 Novelty evidence

Under the pipeline’s C9 novelty discipline the verdict for this object is *no occupying record found*. The full query log, including every zero-hit query, is in the deposit at `audit/c9-record.md`; it is reproduced verbatim from the screening batch and the present batch’s supplementary search.

Summary of channels. OEIS: the sequence, its compressed subcolumn, its even-index variant, its first difference and its partial sums all returned zero hits. The odd-index variant matched ten entries, but every one of them is a zero sequence or a characteristic function (all-zero strings of the kind the database returns for identically zero input), which is a trivial artifact and not an occupying record. Channel availability was established by positive controls before any zero-hit was read: A001835 and A005178 (the unconstrained $3 \times n$ and $4 \times n$ domino-tiling counts) matched, as did A000045. Literature: four channels were queried (arXiv API, Crossref, Mathematics Stack

Exchange, zbMATH Open); there were no direct hits on this constrained-selection object. Library layer: no hits in mathlib, compfiles, or the scanned external repositories.

Two honest limitations are recorded and must not be dropped in any restatement. First, the OpenAlex channel was *unreachable* from the machine used for this search (Cloudflare-fronted hosts timed out; DNS resolution was verified correct against a public resolver, so this is not a DNS-poisoning artifact). Crossref was used as a substitute DOI-level index. Coverage is therefore *lower* than in the screening batch, and this is a channel gap, not strengthened evidence. Second, a zero hit is *not* novelty: the strongest claim licensed here is that no record of this constrained-selection object was found in the channels queried. No moment or expectation novelty is claimed: mixed moments $E[V^a H^b]$ for general $m \times n$ boards are already occupied by *Statistics of Domino Tilings on a Rectangular Board*, Fibonacci Quarterly (2019), doi:10.1080/00150517.2019.12427625 [3]. The $3 \times n$ and $4 \times n$ objects of this family are m -dimensional variants of one another and are not used as independent novelty evidence for each other; the variant partner of the present $3 \times n$ mod-3 track is the $4 \times n$ mod-3 track, which is not cited here as support.

6 Scope and limitations

The kernel-level results concern the pure integer sequences a_2 and b_2 defined above. They do *not* establish that $a_2(n)$ equals the sum of the domino-tiling counts of the $3 \times n$ board over those tilings counted by the number of horizontal dominoes, modulo 3. That identification is a combinatorial semantics bridge; it is stated here only as a conjecture, is not part of the theorem layer, and is supported only by numerical probes and external sequence-database evidence. Accordingly, the results are described throughout as structural theorems about sequences, never as a proved statement about tiling counts. The mod-3 object is exactly the sum over tilings whose horizontal-domino count is congruent to 0 modulo 3; it is not the residue of the total tiling count modulo 3, which is a different object and is not what is formalized here.

Two points about the theorem set must be stated plainly. First, the withdrawn item: the earlier derived recurrence for b_2 was removed because it was degenerate, so this note does not claim a proved order-6 recurrence; the order-6 recurrence is now simply the definition of b_2 , and the three recorded theorems are the entire content of the track. Second, the rework

history recorded in Section 4: the first frozen version defined b_2 from a_2 , which made its recurrence a mere definitional unfolding; that version failed the degeneracy gate and was replaced by an independent definition plus a genuine interleaving bridge between two independent definitions.

The grading of all three theorems is *complete proof* (0 `sorry`, axiom whitelist), and no weakening or extra hypothesis was used. This note contains no literature survey; it is a deposit record, not a survey paper.

Data availability

Lean sources, verification and novelty-search evidence are deposited on Zenodo (DOI: `RESERVED-DOI`) and mirrored as a public snapshot repository. The note is released under CC BY 4.0 and the code under MIT.

Use of generative AI

- *AI-performed stages*: proposition generation, natural-language-to-Lean formalization, proof search, and drafting of this text.
- *Model, node, budget*: pipeline node `anthropic/a6api-main/deepseek-v4.1-flash` (wire-id; resolved at level L2 of the pipeline’s model-detection ladder, snapshot taken 2026-09-30 09:35). Source-task budget: sampling 1 of 8 allowed runs on this track (10 of 32 across four parallel tracks); judgement-class LLM calls 6 of 16 across the batch; Lean formal compilations 4 of 12 on this track (19 of 48 across the batch); LaTeX 0. The numbers are copied from the source task’s `budget.log`.
- *Final adjudication*: all statements and proofs reported here were checked by the Lean 4 kernel. Every proof compiles with no `sorry`, and `#print axioms` reports only `{propext, Quot.sound}` for the three theorems. Statement text was frozen before proving and compared byte-wise afterwards.
- The author reviewed the material and is responsible for all content. AI systems are not listed as authors.

Author contributions

The author determined the research question and the scope fences, verified the semantics of the statements, and is responsible for signing off and for

any deposit or release. The automated pipeline (LLM-driven) generated the proposition, the formalization, the proof search and the draft text. No external release is performed by the pipeline.

References

- [1] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. *CADE-28*, 2021. https://doi.org/10.1007/978-3-030-79876-5_37
- [2] The mathlib Community. The Lean mathematical library. *CPP 2020*, 2020. <https://doi.org/10.1145/3372885.3373824>
- [3] *Statistics of Domino Tilings on a Rectangular Board*. Fibonacci Quarterly, 2019. <https://doi.org/10.1080/00150517.2019.12427625>